



$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

$= 0$ when $z_1 < 0$, so $\partial L / \partial w_1 = \partial L / \partial w_2 = 0$:
 w_1 and w_2 never change, and z_1 stays negative